—NEW THEORY OF THE UNIVERSE—NEWTONIAN IDEAS OVERTHROWN.”11 This was Einstein’s moment of glory.

In the years following this experiment, Eddington’s confirmation of general relativity came under some critical scrutiny. Numerous difficult and subtle aspects of the measurement made it hard to reproduce and raised some questions regarding the trustworthiness of the original experiment. Nevertheless, in the last 40 years a variety of experiments making use of technological advancements have tested numerous aspects of general relativity with great precision. The predictions of general relativity have been uniformly confirmed. There is no longer any doubt that Einstein’s description of gravity is not only compatible with special relativity, but yields predictions closer to experimental results than those of Newton’s theory.

Black Holes, the Big Bang, and the Expansion of Space

Whereas special relativity is most manifest when things are moving fast, general relativity comes into its own when things are very massive and the warps in space and time are correspondingly severe. Let’s describe two examples.

The first is a discovery made by the German astronomer Karl Schwarzschild while studying Einstein’s revelations on gravity in between his own calculations of artillery trajectories at the Russian front during World War I in 1916. Remarkably, just months after Einstein had put the finishing touches on general relativity, Schwarzschild was able to use the theory to gain a complete and exact understanding of the way space and time warp in the vicinity of a perfectly spherical star. Schwarzschild sent his results from the Russian front to Einstein, who presented them on Schwarzschild’s behalf to the Prussian Academy.

Beyond confirming and making mathematically precise the warping that was schematically illustrated in Figure 3.5, Schwarzschild’s work—which has now come to be known as “Schwarzschild’s solution”—revealed a stunning implication of general relativity. He showed that if the mass of a star is concentrated in a small enough spherical region, so that its mass divided by its radius exceeds a particular critical value, the resulting spacetime warp is so radical that anything, including light, that gets too close to the star will be unable to escape its gravitational grip. Since not even light can